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SRI RAMAKRISHNA ENGINEERING COLLEGE, COIMBATORE

(Autonomous Institution, Reaccredited by NAAC with 'A+' Grade Approved by AICTE and Permanently affiliated to Anna University, Chennai)

AUTONOMOUS EXAMINATIONS - NOV'2025

COMMON TO B.E – CSE & M.TECH - CSE

20CS2E22 - DIGITAL FORENSICS

Answer ALL questions

Duration: 3 Hours

Maximum: 100 Marks

PART - A

(7x 1 = 7 Marks)

1. **CO1** After an Incident Responder identifies that a Security incident is in progress, What is The Next Step In The Incident Response Process?
a) Preparation b) Recovery c) Eradication d) Containment
2. **CO1** Which tool is not a Qualified Forensic Duplicate of a Hard Drive?
a) SafeBack b) EnCase c) Mirror Drive d) FTK Imager
3. **CO2** CDRs in mobile forensic stands for _____.
a) Call details records b) Compact disk rewritable
c) Compact disk readers d) Call data records
4. **CO2** Cross Tool validation in digital mobile forensic means _____.
a) use more than one forensic tool in the extraction
b) use an open source tool for validation
c) use of one product tool to confirm the finding of other tools.
d) validation of two or more files that originated from the same source.
5. **CO2** Which section of Information Technology 2000, punishes publication of obscene content in electronic form?
a) 67 b) 69 c) 70 d) 68

6. **CO2** Which storage area keeps cached page elements to speed up reloads?
 a) Cookies b) History c) Cache d) Bookmarks
7. **CO3** What category of software is designed to cause detriment to your computer?
 a) System software b) Network snakes c) Malware d) Bugs

Fill in the blanks: **(3x 1 = 3 Marks)**

8. **CO3** _____ have the capability of spreading itself. It doesn't require the host and human support to spread.
9. **CO2** _____ is a crime committed when someone uses the internet and other technologies to harass another person online
10. **CO3** _____ is the process of deconstructing a product or system to understand its design, structure, and functionality.

PART - B **(5x 3 = 15 Marks)**

11. **CO1** State the need of digital forensics.
12. **CO1** Do you think data recovery is the main role in computer forensics? Justify your answer.
13. **CO2** Describe email forensics.
14. **CO2** The chain of custody is very important for evidence authentication? Justify.
15. **CO3** Classify the types of Rootkits.

PART - C **(5 x 15 = 75 Marks)**

Q.No: 16 Compulsory Question

Q.No: 17 to 22 Answer any FOUR Questions

Compulsory Question:

16. **CO1** i) Explain in detail about Incident Response Methodology and the steps associated with it. (8)
- CO1** ii) How to preserve the digital evidence? Explain it. (7)
17. **CO1** i) Explain the standard procedures of computer forensics. (8)
- CO1** ii) Elaborate about some key network forensics tools. (7)

18. **CO2** i) An employee is suspected of leaking confidential information via a personal webmail account. Report the specific artifacts you would analyze in their web browser to prove their activity. (8)
- CO2** ii) Explain in detail the reconstruction process of web browsing activity. (7)
19. **CO2** i) Illustrate laws against email crimes in relation to the Information Technology Act 2000. (8)
- CO2** ii) Explain the various types of data collected from cell phone. (7)
20. **CO2** i) Explain the key principles of authenticate the evidences. (8)
- CO2** ii) Describe the legal and procedural requirements for giving evidences in court. (7)
21. **CO3** i) Illustrate how to collect information about Linux vulnerabilities. (8)
- CO3** ii) Explain how memory dump is used for malware extraction. (7)
22. **CO3** i) Explain the procedure of extracting malware from Linus systems. (8)
- CO3** ii) Identify the need for reverse engineering in malware analysis. (7)
